* Cayley-Hamilton Th^m: Every square matrix satisfies its own characteristic equ^n.

Problem-1: Verify Cayley-Hamilton th^m for the matrix $A=\begin{pmatrix}1&2\\3&4\end{pmatrix}$.

Sol^n: The characteristic eqn^n of A is

$$|A-\lambda I_2| = 0$$

$$\Rightarrow \left|\begin{pmatrix}1&2\\3&4\end{pmatrix}-\begin{pmatrix}\lambda&0\\0&\lambda\end{pmatrix}\right|=0$$

$$\Rightarrow \begin{vmatrix}1-\lambda&2\\3&4-\lambda\end{vmatrix}=0$$

$$\Rightarrow \lambda^2-5\lambda-2=0 \quad \text{—— (1)}$$

According to Cayley-Hamilton Th^m, we get

$$A^2-5A-2I_2=O \quad \text{where } O=\begin{pmatrix}0&0\\0&0\end{pmatrix}$$

Now $A^2-5A-2I_2=\begin{pmatrix}1&2\\3&4\end{pmatrix}\begin{pmatrix}1&2\\3&4\end{pmatrix}-5\begin{pmatrix}1&2\\3&4\end{pmatrix}-2\begin{pmatrix}1&0\\0&1\end{pmatrix}$

$$=\begin{pmatrix}7&10\\15&22\end{pmatrix}-\begin{pmatrix}5&10\\15&20\end{pmatrix}-\begin{pmatrix}2&0\\0&2\end{pmatrix}$$

$$=\begin{pmatrix}7-5-2&10-10-0\\15-15-0&22-20-2\end{pmatrix}$$

$$=\begin{pmatrix}0&0\\0&0\end{pmatrix}=O$$

Hence the Th^m is verified.

* Find A^{-1} using Cayley-Hamilton Th^m where

$A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$

Sol^n: Characteristic ~~poly~~ eqn^n, $\lambda^2 - 6\lambda - 1 = 0$

$\Rightarrow A^2 - 6A - 1 = 0$

$\Rightarrow A - 6I_2 - A^{-1} = 0$

$\Rightarrow A^{-1} = A - 6I_2$

$= \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} - 6\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}.$

* Do the same for

(i) $B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ (ii) $C = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

CEq: $\lambda^2 - 3\lambda + 1 = 0$

CE: $\lambda^3 - 3\lambda^2 - 5\lambda + 1 = 0$